

HAMZA AHMED

Dokki, Giza, Egypt | 01012370541 | Hamzabatawy@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Computer Science sophomore at Nile University with hands-on industry experience building production-grade RESTful APIs, full-stack web applications, and object-oriented systems. Completed a software engineering internship designing and deploying Java/Spring Boot services against Oracle Database, applying SOLID principles, design patterns, and clean-code practices in a professional team environment. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, and Database Systems, complemented by independently delivered desktop and console applications in Java, C++, and Python. Actively seeking backend, full-stack, or software engineering internship opportunities where technical depth and clean architecture deliver measurable impact.

TECHNICAL SKILLS

Languages: Java · C++ · Python · SQL

Web & APIs: Spring Boot · RESTful API Design · Thymeleaf · HTML · CSS

Databases: Oracle Database · SQL (DDL, DML, Joins, Transactions)

Frameworks & Tools: JavaFX · Git · GitHub · IBM Cloud · Spring MVC

Core Competencies: Object-Oriented Programming · SOLID Principles · Data Structures & Algorithms · Design Patterns · File I/O · Modular Architecture · MVC Design

EDUCATION

Nile University — Cairo, Egypt

Sep 2024 – Present

Bachelor of Science in Computer Science and Information Technology

- Relevant Coursework: Data Structures & Algorithms · Object-Oriented Programming · Java Programming · C++ Programming · Python Programming · Design & Analysis of Algorithms

Stanford International School — Cairo, Egypt

Sep 2021 – Jun 2024

High School Diploma | GPA: 3.85

PROFESSIONAL EXPERIENCE

Alkan Telecom — Solutions & Technology Department

Jul – Aug 2025

Software Engineering Intern · Cairo, Egypt

- Designed and delivered production-ready RESTful API endpoints using Java and Spring Boot, adhering to REST architectural constraints, clean-code standards, and layered service architecture.
- Implemented object-oriented application components applying SOLID principles and GoF design patterns, achieving high cohesion and low coupling across feature modules.
- Authored complex SQL operations against Oracle Database — including multi-table joins, DDL schema management, and DML transactions — to support backend data workflows and reporting pipelines.
- Architected and deployed an enterprise chatbot on IBM Cloud, integrating NLP cloud services to automate end-user query handling and reduce manual support load.
- Built data-driven web pages using Thymeleaf server-side templating within the Spring MVC lifecycle, rendering dynamic model data to client-facing views.

- Collaborated in a team environment using Git and GitHub, managing feature branches, conducting code reviews, and maintaining pull request workflows to ensure repository integrity.

TECHNICAL PROJECTS

Clinic Management System

Java · JavaFX · OOP · MVC Architecture · File I/O | github.com/Hamzahmad6/Clinic-System

- Architected a full-featured desktop application in Java and JavaFX to manage patients, doctors, and appointments, structuring the system around an MVC-inspired pattern with clearly separated Model, Controller, and View layers.
- Engineered appointment workflows — booking, rescheduling, cancellation, and status filtering — with double-booking prevention logic and specialty-based doctor scheduling across a predefined doctor registry.
- Implemented a persistent CSV-based storage layer via a dedicated FileManager class, enabling reliable save-and-load of patient and appointment data on application startup without an external database.

Library Management System

C++ · File I/O · OOP · Modular Design | github.com/Hamzahmad6/LibrarySystemCPP

- Built a console-based library system supporting full CRUD operations on book and user records, organized across logically separated class modules with strong encapsulation and abstraction boundaries.
- Designed a binary and text I/O persistence layer, enabling data durability across application sessions without reliance on an external database or third-party libraries.
- Applied OOP design principles — encapsulation, abstraction, and separation of concerns — resulting in a maintainable and extensible codebase structured for straightforward feature addition.

Tic-Tac-Toe Game

Java · OOP · Console I/O | github.com/Hamzahmad6/TicTacToeGame

- Developed a two-player console game in Java with a clean OOP architecture, separating game state, board rendering, and player interaction into discrete classes for modularity and testability.
- Implemented a win-condition detection engine covering all rows, columns, and diagonals alongside robust input validation, eliminating edge-case failures and invalid move states.

Chemistry Quiz Application

Python · Tkinter · GUI Design | github.com/Hamzahmad6/Python-Chemistry-Quiz

- Developed an interactive GUI quiz application using Python and Tkinter, featuring dynamic question rendering, real-time score feedback, and a randomized engine cycling 10 multiple-choice items per session.
- Structured the codebase with modular functions and logic separation, enabling straightforward extensibility to additional subject domains with minimal refactoring.

GPA Calculator

C++ · HTML · CSS · Algorithm Design

- Formulated a CLI-based GPA calculator in C++ using a weighted-average algorithm, processing credit-hour inputs and applying grade-to-point conversion across an arbitrary number of courses.
- Designed a complementary HTML/CSS web interface as a visual presentation layer, extending accessibility beyond the command-line for non-technical users.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Programming Fundamentals Diploma

Route Academy 2025

C++ Fundamentals · OOP with C++ · Data Structures & Algorithms · Java · Database Fundamentals

AI Career Essentials Certificate

ALX Africa 2024

AI literacy, practical AI tool application, and career readiness for technology-driven environments

LEADERSHIP & CO-CURRICULAR ACTIVITIES

ICPC NU — Operations & Logistics Team

2025–Present

Team Member · International Collegiate Programming Contest, Nile University Chapter

- Coordinated end-to-end event logistics for university-level programming competitions, managing participant registration, venue setup, and multi-round scheduling.
- Collaborated with cross-functional team members to streamline competition operations, contributing to a professionally structured and well-run contest environment.

Student Union Activities Team

2026 – Present

Active Member · Nile University

- Contributed to student engagement initiatives and campus activity coordination, developing organizational and interpersonal skills within a collaborative leadership setting.